HOMEWORK E.2 · EOG REVIEW · NC.8.EE.3 · NC.8.EE.4 · NC.8.G.5 · NC.8.SP.4 · NC.8.G.9

EOG Review: Scientific notation & operations

Targets: EE.3-2, EE.3-1, EE.4-1, EE.4-3, EE.3-4, G.5-3, EE.4-6, EE.3-7, SP.4-4, EE.4-7, G.9-10

Name _____ Date _____ Block _____

Remember: Scientific notation: $a \times 10^n$ with $1 \le a < 10$. Big number → positive exponent. Less than 1 → negative exponent. **Multiply**: multiply the numbers, **add** the exponents. **Divide**: divide the numbers, **subtract** the exponents. Example: $(3 \times 10^4)(2 \times 10^5) = 6 \times 10^9$.

SCORE 3.0 Approaching · the basics

Targets: EE.3-2, EE.3-1, EE.4-1, EE.4-3, EE.3-4, G.5-3

1 Write in standard form: 3.4×10^{-3} EE.3-2

2 Which shows 58,000,000 in scientific notation? EE.3-1

(A) 5.8×10^7
(B) 58×10^6
(C) 5.8×10^6
(D) 5.8×10^{-7}

3 Multiply. Write the answer in scientific notation. EE.4-1,EE.4-3

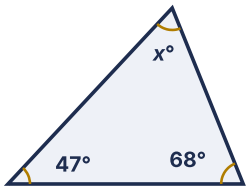
$(6 \times 10^5)(5 \times 10^3)$

4 Circle >, <, or =: $4 \times 10^{-3} \circ 4 \times 10^{-5}$ EE.3-4

(>) (<) (=)

Why? _____

5 Find x. REVIEW 4.5 G.5-3



SCORE 4.0 Proficient · on target

Targets: EE.4-6, EE.3-7, SP.4-4

6 A hard drive holds 8×10^{11} bytes. One photo uses 4×10^{6} bytes. How many photos fit on the drive? EE.4-6

7 **Kayla** made a mistake. Circle it. Explain it. Fix it. EE.3-7

KAYLA'S WORK

$0.0072 = 7.2 \times 10^{3}$

8 What percent of 8th graders play a sport? REVIEW 7.3 SP.4-4

Plays a sport?	Yes	No	Total
Grade 7	28	22	50
Grade 8	36	24	60
Total	64	46	110

SCORE 5.0 Mastery · go deeper

Targets: EE.4-7, G.9-10

9 About 8×10^{9} people live on Earth. Each person drinks about 2 liters of water a day. About how many liters is that each day? Explain each step. EE.4-7

I _____ the numbers. I got _____. In scientific notation it is _____ liters.

10 A cylinder has radius 3 cm. Its radius is doubled to 6 cm. The height stays the same. What happens to the volume? Explain. REVIEW 9.3 G.9-10

The volume becomes _____ times as big because _____.